

Design of Experiments on Networks for Autoregressive Models

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Abstract

We investigate how to design experiments under an autoregressive model, where the response of an experimental unit is affected by the response of others that it is connected to, according to an autoregressive parameter as well as the strength of connections between the experimental units, specified by a known adjacency matrix.

We derive the Fisher information for this model, and as the optimal design depends on the unknown parameters, we present (pseudo-)Bayesian optimal designs for point priors (locally optimal designs) and simple priors.

We extend the model to include blocking, and indirect treatment effects, and present optimal designs for a wide range of models useful on experiments on networks, including spatial models.

We give examples in crop experiments, marketing experiments on social networks, crossover trials, and spatial experiments, but the ideas and methods can readily be generalised to many designed experiments on networks.

Keywords: Experimental Design; Mixed Regressive Spatial Autoregressive Model; Spatial Lag Model

1 Introduction

Experimental design often involves finding an allocation of treatments to experimental units to maximise the information about the treatment effects under consideration. Often an experimental design makes the Stable Unit Treatment Value Assumption

(SUTVA)([1]), that the response of one experimental unit does not depend on the treatment applied to other experimental units. Whilst this assumption often holds, in many situations experimental units may be connected in a network, that is they are affected by the responses of other experimental units they are connected to. In some experiments we may even wish to investigate the connections themselves. Some motivating examples are:

- a crop experiment to assess the effect of fertilisers on biodiversity; we apply several treatments (fertilisers) to experimental units (plots) which are geographically separated, and measure the response (biomass of insects present) in each experimental unit after one year. We expect that the response in one plot may be determined by the response in other plots, as insects may move from one plot to the other; geographical distance between the plots is likely to play a part. It is of interest in this kind of experiment to ensure that the experiment is efficient as possible at estimating the effect of the fertilisers, as well as (possibly) the effect of the geographic separation.
- an experiment where different adverts (treatments) are sent to different potential customers (experimental units). We measure a response, which is the amount spent on the product after advertising. The amount spent by each potential customer may be influenced by other customers who they are in contact with according to a social network. In this experiment, we might want to determine which advert is most effective, allowing for this peer-effect in the social network.
- an experiment on a London borough where different advertising regimes (treatments) are used in different wards (experimental units) to promote recycling. We measure the amount of waste recycled in the ward after the intervention. In this experiment, we might want to determine how effective the advertising is, knowing that people travel between wards and may see the advertising from other wards. See Section 3.4 for a fuller description.

Experimental designs allowing for connections between experimental units have mostly been studied in the context of agricultural experiments. These connections include those between neighbours in a plot (e.g. Williams [2], Azais et al. [3]), and between neighbouring blocks (e.g. Pearce [4]), but, perhaps due to computational limitations at the time, simple models were used where relationships were considered only with the assumption that immediate neighbours were connected, or in some kind of regular lattice pattern.

Recent work ([5]) has relaxed this assumption of simple connectivity, and showed how a general network structure could be considered, motivated by social networks. That work introduced the linear network effects model, and considered how to design an experiment where the *treatment* assigned to an experimental unit affects its neighbours, and in this work we describe how to design experiments when the *responses* between neighbours are correlated.

1.1 Autoregressive model considered

In this paper we investigate how to design experiments under an autoregressive model, where the response of one experimental is affected by the response of those it is

connected to, according to an autoregressive parameter as well as the strength of connections between the experimental units, specified by adjacency matrix. We derive the Fisher information for this model, and present optimal designs for networks of experimental units.

For simplicity of exposition, we mainly consider the model often called the Mixed Regressive Spatial Autoregressive model (MRSAR), although also called the Spatial Lag model, where the response of each node depends on the response of each connected unit as follows:

$$Y_i = \mu + \tau_{t(i)} + \rho \sum_{k=1}^n A_{ik} Y_k + \epsilon_i, \quad (1)$$

where $i \in \{1, \dots, n\}$ is the experimental unit, $t(i) \in \{1, \dots, m\}$ is the treatment given to the i -th experimental unit, τ_j is the treatment effect of the j -th treatment, $j \in \{1, \dots, m\}$, ρ is an autoregressive parameter, A is the adjacency matrix, a $n \times n$ matrix where $0 \leq A_{ik} \leq 1 \forall i \neq k$, $A_{ii} = 0 \forall i$, and we assume that ϵ_i are independent and identically normally distributed with mean 0 and constant variance σ^2 .

The adjacency matrix A can either be weighted or unweighted, and can be directed or undirected. This means that a wide range of models can be encompassed within this framework. For example, if A is an unweighted (binary) matrix, experimental units can either be connected or not such as in a social network; with a weighted adjacency matrix A can represent a distance metric, which could represent a spatial model. We show that this model can trivially be expanded to include blocks and also propagation of indirect treatment effects through the network, to allow a much wider class of models that could be used to design many experiments.

The remainder of this article is set out as follows. In Section 2 we review previous work for designing experiments on networks and spatial models; in Section 3 we discuss the MRSAR model under question, and derive the Fisher information. We demonstrate optimal designs through some examples in Section 4 and make conclusions in Section 5.

2 Previous work

We consider experiments where experimental units are connected in some way to some other experimental unit(s), and that the response of an experimental unit depends in some way on either the response or the treatment given to neighbouring units. This has been investigated in various different ways in the literature.

2.1 Design for field experiments with correlations

Williams [2] considers designs where the errors between plots are correlated according to a stationary Markov process with its neighbour. Pearce [4] considers a model for biological experiments where experimental units in a block exhibit a “remote effect” on experimental units in the same block, but not on units in different blocks, and presents an example of pollination in apple trees. Martin [6] investigate optimal designs for rectangular arrays under a linear model for where the errors are correlated and demonstrates the efficiency of some designs for small rectangular arrays.

If an experimental unit is affected by a treatment applied to a neighbouring experimental unit, Azais et al. [3] present a construction method for neighbour-balanced designs, and show that they are efficient under various models with either left-effects, right-effects, or both.

There is a large body of work (see Verdooren et al. [7], Section 7 for a review) dealing with spatial analysis in field experiments. Generally this work assumes that in a field experiment, some regions will produce larger or smaller responses due to, for example, soil conditions within the field. Seminal work in this area includes Patterson et al. [8] where α -designs are introduced, a method of construction for resolvable Balanced Incomplete Block designs such that a design is produced where treatment effects can be estimated better allowing for spatial variation. We do not consider these models here, as we are concerned with how relationships between responses affect designs, rather than underlying spatial variation.

2.2 Design of experiments on networks

One of the first paper to consider experimental design for units with a network structure was [5]. This considered small example networks from social networks, and also agriculture, crossover trials, etc, and presented a framework to construct optimal designs where the effect of a treatment propagated to neighbours of the experimental unit it was originally applied to. This model was developed in [9], and subsequent work, including models incorporating blocks [10], algorithms for design, applications to field experiments, and using isomorphisms to improve designs for networks and other networks [11].

Approaching a similar problem from a Causal Inference framework, Eckles et al. [12] discusses issues in design and analysis when allocating treatments to experimental units on networks; two treatments are considered and the response of each unit depends on its own treatment, and the sum of responses to its neighbours. Results about bias and variance are derived and simulation results presented, with more consideration given to minimising the bias. [13] generalises this to more complicated models and discusses restrictions on randomisations that allows for properties such as bias and variance of optimal designs to be found, including under model misspecification. This general idea is carried forward in [14] where testing for the existence of an interference effect is considered by proposing methods for treatment assignments on a network which has been divided into halves. Bowers et al. [15] provides some results from simulation experiments showing the importance of designs in social networks, noting that “prior information regarding treatment propagation should be incorporated into the experimental design. . . Information about treatment effects is not maximised when we assign half the units to treatment and half to control”.

A general class of networks called the General Additive Network Effect mode, is considered in Bui et al. [16]; in that model a wide class of experiments is studied, which includes the Local Aggregate model, which is very similar to that studied in this article. That article focuses on experimental design for a POW-DEG model, and finds many of the same general conclusions for a different model, in a different way to what follows, although demonstrates the difficulty of experimental design on these non-linear models.

3 Experiments on Networks

We wish to consider in this work how we would design an experiment on a network where we knew the strength of connection between experimental units (i.e. the adjacency matrix A is known).

We argue that there are three general considerations when designing experiments on networks. Consider that experimental units F and G are connected.

1. If we give a treatment to F , G 's response may be modified by the treatment given to F (an indirect treatment effect).
2. If we give a treatment to F , G 's response may be modified because F 's response is modified by the treatment (Autoregressive effect)
3. F and G are connected because there is some similarity between them, so their responses are similar (Blocking, or Spatial Analysis)

We note that (1) was introduced in [5]; both (1) and (3) were considered in [10]. In this current work we introduce (2), and describe how we can easily combine it with (1) and (3).

In practice, these three considerations can be difficult to separate conceptually. Consider our second motivating example on advertising a product in a social network: if we wish to assess the effectiveness of advertising exercise equipment, we might consider whether the reason people were connected via a social network is that they were members of a sports club. Members of a sports club may be more likely to buy exercise equipment (a blocking effect) irrespective of their friends' behaviour (an autoregressive effect). Care will need to be taken by experimental designers for each experiment considered.

3.1 A model for network experiments

A network $G = (N, E)$ is described by a directed or undirected graph, a collection of nodes N and edges $E \subseteq (N \times N)$, where the nodes represent experimental units on each of which we apply some treatment. The edges represent relationships between the experimental units. We have $|N| = n$ experimental units, and m treatments. The strength of the relationship between experimental units is specified by the $n \times n$ adjacency matrix A where $A_{ik} \geq 0$ if and only if i and k are connected and $A_{ik} = 0$ otherwise. We consider simple graphs without self loops and impose the restriction $A_{ii} = 0$.

We assume that the experimental units are grouped into B disjoint blocks. We combine these different considerations into one model,

$$Y_i = \mu + \tau_{t(i)} + z_{b(i)} + \sum_{k=1}^n A_{ki} \gamma_{t(k)} + \rho \sum_{k=1}^n A_{ki} Y_k + \epsilon_i, \quad (2)$$

where $i \in \{1, \dots, n\}$ is the experimental unit, $t(i) \in \{1, \dots, m\}$ is the treatment given to the i -th experimental unit, τ_j is the treatment effect of the j -th treatment, $j \in \{1, \dots, m\}$, $z_{b(i)}$ is a block effect caused by the experimental unit being in block $b(i) \in \{1, \dots, B\}$, ρ is an autoregressive parameter, $0 \leq \rho \leq 1$, A is an $n \times n$ adjacency matrix where $0 \leq A_{ik} \leq 1 \quad \forall i \neq k$, $A_{ii} = 0 \quad \forall i$, and we assume that ϵ_i are independent

and identically normally distributed with mean 0 and constant variance σ^2 . Thus the response of the i -th experimental unit now depends on

- a grand mean μ
- a treatment effect corresponding to the treatment given directly to the experimental unit;
- a block effect depending on the block that the experimental unit is in;
- the sum of the indirect treatment effect corresponding to treatments given to neighbours of the experimental unit, weighted for the strength of the influences of those neighbours;
- the sum of the autoregressive effects due to the experimental units response being modified due to the response of its neighbours;
- an error term.

3.2 The Mixed Regressive Spatial Autoregressive model

For simplicity of exposition, we now assume $\mathbf{z} = \mathbf{0}$ and $\boldsymbol{\gamma} = \mathbf{0}$, i.e. there are no block effects or indirect treatment effects. This corresponds to the model introduced above as model (1).

Spatial models are frequently used in econometrics to incorporate spatial structure into regression models, e.g. [17]. Various methods of estimation exist, which we do not consider in this paper, but refer the reader to [18] for discussion on estimation. This model (1) is often known as the MRSAR model in econometrics.

We assume that each experimental unit receives exactly one treatment, and that we have unstructured treatments. We can write this model in vector form as

$$\mathbf{Y} = X\boldsymbol{\beta} + \rho A^T \mathbf{Y} + \boldsymbol{\epsilon}$$

where X is the $n \times (m+1)$ design matrix $X = (\mathbf{1} \quad \mathbf{u}_1 \quad \mathbf{u}_2 \dots \quad \mathbf{u}_m)$ where $\mathbf{1}$ is the length n vector with all elements 1, and u_{ij} is the vector with all elements zero except $u_{ij} = 1$ wherever unit i is given treatment j , and $\boldsymbol{\beta} = (\mu \quad \boldsymbol{\tau})^T$.

We can rearrange this as

$$\begin{aligned} \mathbf{Y} - \rho A^T \mathbf{Y} &= X\boldsymbol{\beta} + \boldsymbol{\epsilon} \\ \mathbf{Y} &= (I - \rho A^T)^{-1} (X\boldsymbol{\beta} + \boldsymbol{\epsilon}) \end{aligned}$$

wherever the inverse $(I - \rho A^T)^{-1}$ exists. Let $K = K(\rho) = (I - \rho A^T)^{-1}$; if the inverse does not exist, we let K be a generalised inverse $K = K(\rho) = (I - \rho A^T)^-$ and adapt what follows appropriately (the inverse exists for most applications on real networks; lack of existence of the inverse corresponds to not being able to estimate the elements of $\boldsymbol{\beta}$ uniquely, such that the information function value we find later is zero.)

Thus $Y \sim N(KX\boldsymbol{\beta}, K(\sigma^2 I)K^T) = N(KX\boldsymbol{\beta}, \sigma^2 KK^T)$

Remembering σ^2 is a constant, and writing $\boldsymbol{\theta} = (\mu \quad \boldsymbol{\tau}^T \quad \rho)^T$, a length $(m+2)$ vector, $M = KX\boldsymbol{\beta}$, and $\Sigma = \sigma^2 KK^T$ we can show (see [19]) that the Fisher Information matrix is an $(m+2) \times (m+2)$ matrix where

$$I_{i,j} = \frac{\partial M^T}{\partial \theta_i} \Sigma^{-1} \frac{\partial M}{\partial \theta_j} + \frac{1}{2} \text{tr} \left(\Sigma^{-1} \frac{\partial \Sigma}{\partial \theta_i} \Sigma^{-1} \frac{\partial \Sigma}{\partial \theta_j} \right).$$

As $\Sigma^{-1} = \frac{1}{\sigma^2} (KK^T)^{-1}$, we can rewrite this as

$$I_{i,j} = \frac{1}{\sigma^2} \frac{\partial M^T}{\partial \theta_i} (KK^T)^{-1} \frac{\partial M}{\partial \theta_j} + \frac{1}{2\sigma^4} \text{tr} \left((KK^T)^{-1} \frac{\partial \Sigma}{\partial \theta_i} (KK^T)^{-1} \frac{\partial \Sigma}{\partial \theta_j} \right).$$

We note that $\frac{\partial \Sigma}{\partial \theta_i} = 0$ for $i \in \{1 \dots (m+1)\}$, and that $\frac{\partial M}{\partial \theta_1} = K\mathbf{1}$, $\frac{\partial M}{\partial \theta_j} = K(X_{1j} \ X_{2j} \ \dots \ X_{nj})^T$ for $j = 2 \dots (m+1)$. Further,

$$\begin{aligned} \frac{\partial M}{\partial \theta_{n+2}} &= \frac{\partial M}{\partial \rho} = \left(\frac{\partial K}{\partial \rho} \right) X\beta \\ &= \left(\frac{\partial (I - \rho A)^{-1}}{\partial \rho} \right) X\beta \\ &= -(I - \rho A)^{-1} \left(\frac{\partial (I - \rho A)}{\partial \rho} \right) (I - \rho A)^{-1} X\beta \\ &= -(I - \rho A)^{-1} (-A) (I - \rho A)^{-1} X\beta \\ &= (I - \rho A)^{-1} A (I - \rho A)^{-1} X\beta \\ &= KAKX\beta \end{aligned}$$

where the third line can be seen by noting that if K is an invertible matrix, then the derivative of its inverse $\frac{\partial (K^{-1})}{\partial \rho} = -K^{-1} \frac{\partial K}{\partial \rho} K^{-1}$. Finally,

$$\begin{aligned} \frac{\partial \Sigma}{\partial \theta_{m+2}} &= \frac{\partial}{\partial \rho} (\sigma^2 KK^T) = \sigma^2 \left[K \frac{\partial K^T}{\partial \rho} + \frac{\partial K}{\partial \rho} K^T \right] \\ &= \sigma^2 (K(KAK)^T + (KAK)K^T) \\ &= \sigma^2 (K(K^T A^T + AK)K^T) \end{aligned}$$

so

$$\begin{aligned} I_{m+2,m+2} &= I_{\rho,\rho} = \frac{1}{\sigma^2} \frac{\partial M^T}{\partial \rho} (KK^T)^{-1} \frac{\partial M}{\partial \rho} + \frac{1}{2\sigma^4} \text{tr} \left((KK^T)^{-1} \frac{\partial \Sigma}{\partial \rho} (KK^T)^{-1} \frac{\partial \Sigma}{\partial \rho} \right) \\ &= \frac{1}{\sigma^2} (KAKX\beta)^T (KK^T)^{-1} KAKX\beta + \frac{1}{2} \text{tr} \left([(KK^T)^{-1} K[K^T A^T + AK]K^T]^2 \right), \\ &= \frac{1}{\sigma^2} (AKX\beta)^T (AKX\beta) + \frac{1}{2} \text{tr} \left((KK^T)^{-1} KK^T [K^T A^T + AK]^2 \right), \\ &= \frac{1}{\sigma^2} (AKX\beta)^T (AKX\beta) + \frac{1}{2} \text{tr} \left([K^T A^T + AK]^2 \right), \end{aligned}$$

where the second term in the third line simplifies as $\text{Tr}(M_1 M_2) = \text{Tr}(M_2 M_1)$ for arbitrary matrices M_1, M_2 .

Putting this altogether gives us

$$I = \frac{1}{\sigma^2} \begin{pmatrix} n & n_1 & n_2 & \dots & n_m & \mathbf{1}^T AKX\beta \\ n_1 & n_1 & 0 & \dots & 0 & (X_{11} \ X_{21} \ \dots \ X_{n1})AKX\beta \\ \vdots & \vdots & \vdots & \ddots & \vdots & \\ n_j & 0 & \dots & \dots & 0 & (X_{1j} \ X_{2j} \ \dots \ X_{nj})AKX\beta \\ \vdots & \vdots & \vdots & \ddots & \vdots & \\ n_m & 0 & 0 & \dots & n_m & (X_{1m} \ X_{2m} \ \dots \ X_{nm})AKX\beta \\ \mathbf{1}^T AKX\beta & \dots & \dots & \dots & \dots & (AKX\beta)^T(AKX\beta) + \frac{\sigma^2}{2} \text{tr}([K^T A^T + AK]^2) \end{pmatrix}. \quad (3)$$

This can be represented as the block matrix

$$I = \frac{1}{\sigma^2} \left(\begin{array}{c|c} X^T X & X^T AKX\beta \\ \hline X^T AKX\beta & (AKX\beta)^T(AKX\beta) + \frac{\sigma^2}{2} \text{tr}([K^T A^T + AK]^2) \end{array} \right).$$

where the left column/top row refers to information about the parameters β and the right column/bottom row to that about ρ .

3.3 Extending the Fisher Information matrix for a general model

We now show that as we have found the Fisher information matrix for model (1), we can find the Fisher information for the more general model including block and indirect treatment effects Returning to the more general model (2):

$$Y_{ij} = \mu + \tau_{t(i)} + z_{b(i)} + \sum_{k=1}^n A_{ki} \gamma_{t(k)} + \rho \sum_{k=1}^n A_{ik} Y_k + \epsilon_{ij},$$

We can write this model in vector form as

$$Y = X(\mu \ \tau^T)^T + Z\mathbf{b} + A^T X \gamma + \rho A^T Y + \epsilon$$

where now Z is the block incidence matrix such that $Z_{ij} = 1$ if experimental unit i is in block j , and $Z_{ij} = 0$ otherwise, and all other terms are as before.

We define our vector of unknown parameters β in this model as

$$\beta = (\mu \ \tau^T \ \mathbf{b}^T \ \gamma^T)^T,$$

and as the unknown terms τ and $\mathbf{b}$ only exist in the model as linear terms, we can deduce that our information matrix is now given by the block matrix

$$I = \frac{1}{\sigma^2} \begin{pmatrix} X^T X & X^T Z & X^T A X & X^T A K X \beta \\ & Z^T Z & Z^T A X & Z^T A K X \beta \\ & & (A X)^T A X & (A X)^T A K X \beta \\ & & & (A K X \beta)^T (A K X \beta) + \frac{\sigma^2}{2} \text{tr}([K^T A^T + A K]^2) \end{pmatrix}.$$

where only the upper diagonal terms are shown as the matrix is symmetric. Throughout the rest of this article, we mostly consider the simpler model (1). Although the general model is more flexible, the block and indirect treatment effects have been studied in [5] and [10] and in this current article for clarity of exposition we seek to see how the non-linear autoregressive term affects the design.

3.4 Finding an optimal design

Finding an optimal design depends on choosing a treatment allocation $\mathbf{t} = (t(1), t(2), \dots, t(n))^T$ to maximise some function of the information matrix defined in (3), which is equivalent to minimising some function of the variance-covariance matrix corresponding to the uncertainty in estimating unknown parameters, which may be of interest for a particular experiment.

We assume in this paper that we wish to use maximum likelihood estimation. In general, we may be interested in estimating τ , ρ , γ or sometimes a combination of these.

We note that our information matrix I (defined in (3)) depends on our design $\mathbf{t}$ and unknown parameters β and ρ , so we write it $I(\beta, \rho, \mathbf{t})$.

Estimating ρ

We wish to estimate ρ as precisely as possible, which corresponds to minimising $\text{Var } \rho$. This corresponds to minimising

$$\psi_\rho(\beta, \rho, \mathbf{t}) = I_{m+2, m+2}^{-1}(\beta, \rho, \mathbf{t})$$

i.e. the bottom right element of the inverse of the information matrix.

Estimating τ

We seek to minimise the average variance of all pairwise differences of treatment effects

$$\psi_\tau = \frac{2}{m(m-1)} \sum_{s=1}^{m-1} \sum_{s'=s+1}^m \text{var} \left(\widehat{\tau_s - \tau_{s'}} \right),$$

We do this in the following way

Definition. Let $s(\alpha_1, \alpha_2)$ be a contrast vector formed in the following way:

1. Form a vector of $2m + 2$ zeroes, corresponding to the mean μ , the m treatment effects, τ , and ρ .

2. Let the α_1 -th and α_2 -th elements be 1 and -1 respectively, corresponding to the particular effects we wish to estimate in a given contrast.
3. Set the $(m+1)$ -th element to be 0, without loss of generality, allowing us to estimate the treatment effects uniquely.

The resulting vectors of length $(m+2)$ are pre- and post-multiplied by the $(m+2) \times (m+2)$ matrix I^{-1} to allow us to calculate ψ_τ as

$$\psi_\tau(\boldsymbol{\beta}, \rho, \mathbf{t}) = \sum_{v=2}^m \sum_{q=v+1}^{m+1} \mathbf{s}^T(v, q) I^{-1}(\boldsymbol{\beta}, \rho, \mathbf{t}) \mathbf{s}(v, q) ,$$

where in general our information matrix and therefore the value of our design criterion, ψ_τ , depends on the (usually unknown) values of $\boldsymbol{\beta}$ and ρ .

It is also possible to want to estimate both τ and ρ for the same experiment, so we could for example define a compound criterion such as $\psi_\alpha = \alpha\psi_\rho + (1 - \alpha)\psi_\tau$ for $0 \leq \alpha \leq 1$; we do not deal with this here.

3.5 Bayesian Design

Our model (1) is non-linear in $\boldsymbol{\beta}$ and ρ , so the information matrix and thus the information criteria values ψ_τ and ψ_ρ depend on these unknown parameters, which we try to estimate.

We approach this via (pseudo-) Bayesian experimental design. We assume a prior distribution for our unknown parameters $p(\boldsymbol{\beta}, \rho, \sigma^2)$ so that our optimal designs are given by

$$\psi_\tau(\mathbf{t}) = \int_{\sigma^2} \int_{\boldsymbol{\beta}} \int_{\rho} \psi_\tau(\boldsymbol{\beta}, \rho, \mathbf{t}) p(\boldsymbol{\beta}, \rho, \sigma^2) d\boldsymbol{\beta} d\rho d\sigma^2 ,$$

and similarly for ψ_ρ . We then seek to find the design $\mathbf{t}$ which minimises this integral.

In general, the integral is computationally intractable, so various techniques are used to approximate it, such as quadrature or Monte Carlo techniques. In the examples that follow, we adopt a simple quadrature for simplicity of exposition.

3.6 Algorithms and software

An R package to find designs for networks is available at github.com/bmp22/networkDesign. Scripts are provided to allow the reader to reproduce the results in this paper.

For the small networks described in this paper, we are able to use an exhaustive search to find the optimal design. Other techniques and algorithms are described more fully in [11], and are the subject of further research.

4 Results

We ran an exhaustive search for various networks, available as part of the R Package at github.com/bmp22/networkDesign. These designs are social networks (Examples 1, 2 and 3), crop trials (Examples 4 and 5), a crossover trial (Example 6), taken from [5]. We also consider some regular lattice designs.

Network	Area	n	Comment
Example 1	Social network	10	A small sparsely connected social network
Example 2	Social network	10	A small densely connected social network
Example 3	Social network	20	A larger densely connected social network
Example 4	Agricultural experiment	12	A small spatial network with left and right neighbour effects
Example 5	Agricultural experiment	15	An irregularly shaped network with neighbour effects
Example 6	Crossover trial	15	A directed network representing a crossover trial on 4 patients over four periods, with one planned absence.

Table 1: Summary of networks considered when finding optimal designs. See [5] for full details.

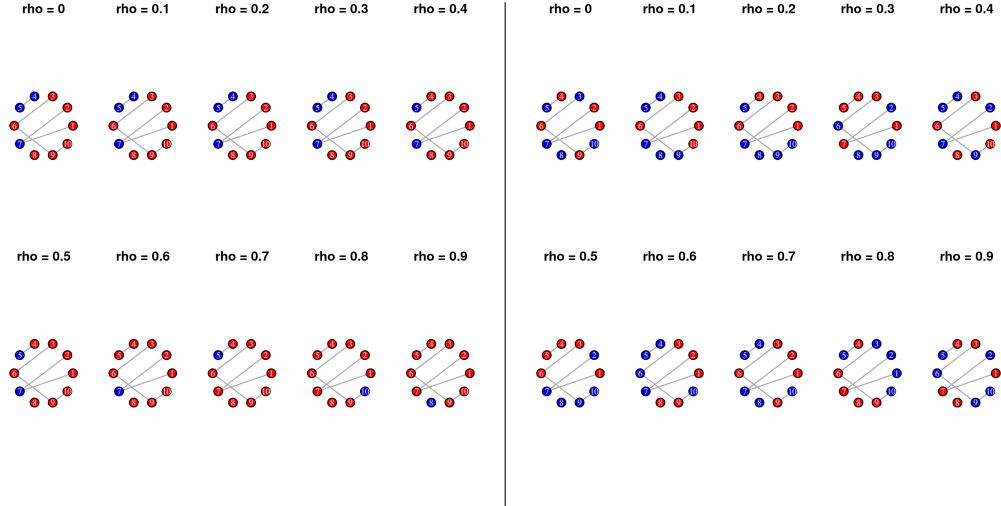


Fig. 1: Optimal design for two treatments on example 1 network from [5] for estimating (left) ψ_ρ and (right) ψ_τ

A summary of the networks studied forms Table 1. Throughout this section we find optimal designs under model (1), and demonstrate for $m = 2$ treatments.

4.1 Locally Optimal designs

As the Fisher Information depends on the unknown parameters β, ρ, σ^2 , we arbitrarily chose designs which are locally optimal, i.e. the design which is optimal for a particular chosen value of these parameters that we assume are unknown before doing the experiment. This is equivalent to choosing a point prior for the unknown parameters.

We chose point priors of $\beta_0 = \mu = 30, \beta_1 = \tau_1 = 10, \beta_2 = \tau_2 = 0, \sigma^2 = 1$ and present results for $\rho = 0, 0.1, 0.2, \dots, 0.9$.

Results are shown in Figures 1, 2, 3, and in the Appendix. Treatment 1 is shown in red, treatment 2 in blue. Designs for further networks are shown in the appendix.

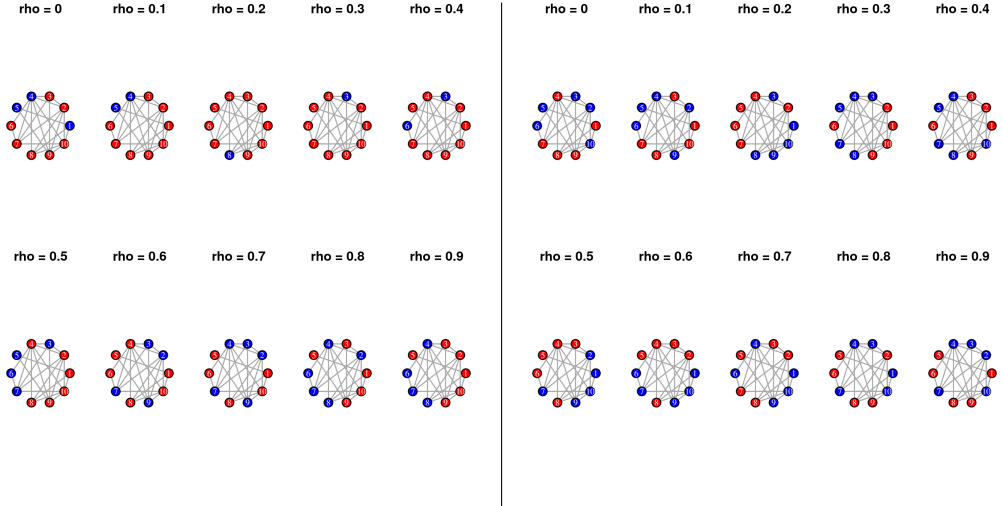


Fig. 2: Optimal design for two treatments on example 2 network from [5] for estimating (left) ψ_ρ and (right) ψ_τ

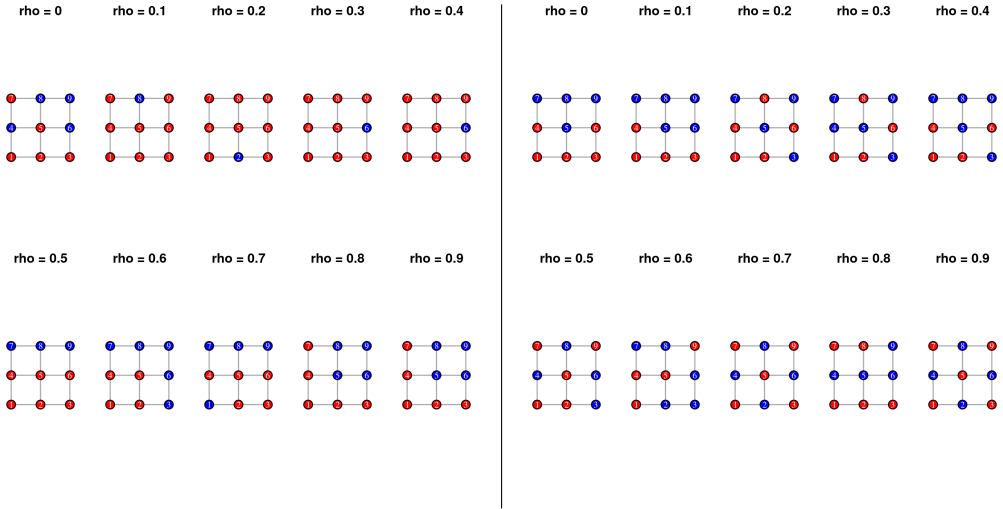


Fig. 3: Optimal design for two treatments on 3×3 lattice network network for estimating (left) ψ_ρ and (right) ψ_τ

Although the exact optimal treatment allocation is difficult to predict, we see some patterns. Although we have considered estimating ρ and τ separately for our final criterion, the two cannot truly be decoupled: if we know ρ we can more accurately estimate τ and vice versa. Designs thus seek to do several things:

- For estimating τ it is beneficial to have as balanced a design as possible- as far as possible an equal number of each treatment.
- In our experiments the prior includes $\tau_1 = 10, \tau_2 = 0, \sigma^2 = 1$; when we have a smaller autoregressive effect in the network, we would rather have larger values of our response variable for each node. This allows us to distinguish the autoregressive effect as it propagates through the network and makes all the responses larger, and makes the autoregressive effect easier to separate in comparison to the error, which here is assumed small compared to the response. We would thus rather see more nodes assigned the treatment with larger effect, treatment 1, with a red colour. The more connected nodes in the network should also be coloured red.
- When ρ approaches 1, for a highly connected network if all nodes were the same colour, it is not possible to distinguish the autoregressive effect from the treatment effect. Thus again the balance between nodes becomes more important for much higher values of ρ
- Red nodes also are more likely to be highly connected nodes as this also ensures the response propagates through the network and makes it easier to estimate ρ .
- We only show one optimal design here. It is clear that there can in general be many designs which are optimal. See [11] for some discussion.
- Despite noting these general principles, assessing heuristically the balance between them is very hard, and for any particular network and prior a numerical search will be necessary.

4.2 Simple Bayesian Priors

It is usual for an experimenter to have some idea about the value of the unknown parameters in the form of a prior, which will inform on the optimal experimental design. To demonstrate this, we considered example 2 from [5], and specified two priors.

- Prior 1: $\mu = 2, \tau_2 = 1, \tau_1 = 0$, and $\rho \sim N(0, 0.25^2), |\rho| < 1$
- Prior 2: As prior 1, except now the prior was $\rho \sim U[0, 1]$

We approximated prior 1 using an 11 point Gauss Hermite quadrature (with no point such that $|\rho| \geq 1$), and prior 2 using an 11 point discrete uniform distribution and found the optimal designs under model (1). The results are shown as Figure 4.

We see similar patterns to the locally optimal design, and although the exact designs shown are unremarkable, we show that we can adapt our methodology to more realistic experiments where our treatment effects and autoregressive parameter have some prior on them, but we would like to more precisely estimate them.

4.3 Spatial Networks

Up to now, our examples have used adjacency matrices A which are binary, but nothing in our theory requires this to be so. Here we generate 15 nodes uniformly at random in $[0, 1] \times [0, 1]$ and let the adjacency matrix represent the reciprocal of the Euclidean distance between the points, such that there is a stronger connection between points which are closer to each other. We then proceed as before and find the optimal treatment allocation for finding ψ_τ and ψ_ρ .

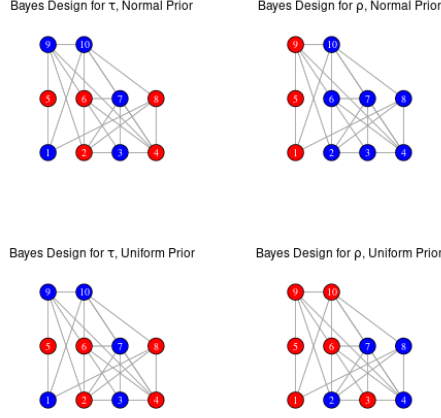


Fig. 4: (pseudo)-Bayesian designs for two simple priors on ρ , (top) $\rho \sim N(0, 0.25^2)$, $|\rho| < 1$, (bottom) $\rho \sim U[0, 1]$ under model (1) for estimating (left) ψ_τ and (right) ψ_ρ

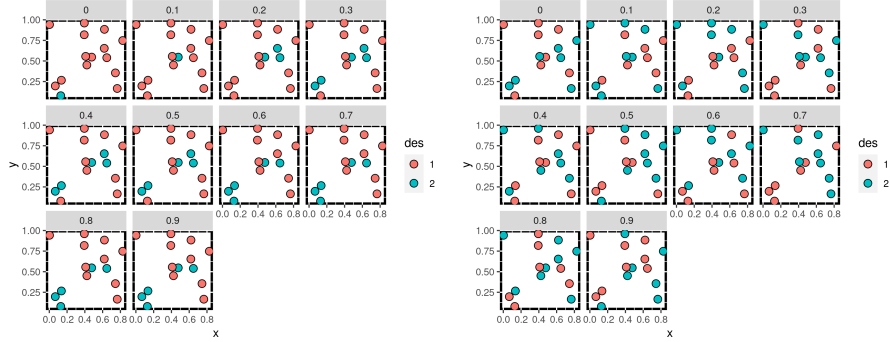


Fig. 5: Optimal spatial designs for points randomly chosen in $[0, 1] \times [0, 1]$ for varying ρ . (Left) shows optimal designs for ψ_ρ , estimating autoregressive parameter. (Right) shows optimal designs for ψ_τ , estimating treatment difference $\tau_2 - \tau_1$.

As before, we arbitrarily chose point priors of $\beta_0 = \mu = 0 = 20$, $\beta_1 = \tau_1 = 10$, $\beta_2 = \tau_2 = 0$, $\sigma^2 = 1$ and presented results for $\rho = 0, 0.1, 0.2, \dots, 0.9$. Treatment 1 is shown in red, treatment 2 in blue, and results are presented as Figure 5

Similar features to before are found- we seek to balance the number of each treatment as far as possible. In addition, a greater number of red treatments are preferred to allow us to have larger responses more effectively distinguish τ and ρ in the presence of normal random error.

We see an interesting pattern more clearly in this spatial model for estimating ρ . Clusters in the design seem to appear for higher assumed ρ ; clusters with all the same

treatment will allow us to estimate the autoregressive parameter well, and having different coloured clusters at a greater distance which affect each other less allows us to estimate the different treatment effects well, which informs on the estimation of ρ .

4.4 Efficiency

Although we have set out how to find optimal designs, we have not hereto shown whether an optimal design is of importance for real experiments. Are sub-optimal designs only slightly worse than optimal designs, in which case there may be no point in designing network experiments?

One measure of the effectiveness of the design is to compare the efficiency of designs. We let our optimal designs for ρ and τ to be $\psi_\rho^*(\beta, \rho, t)$ and $\psi_\tau^*(\beta, \rho, t)$. We define the efficiency for estimating τ for any particular design t for unknown β, ρ to be to be $\psi_\tau(\beta, \rho, t)/\psi_\tau^*(\beta, \rho, t)$ and similarly for estimating ρ . For the criteria we consider, this is equivalent to the ratio of the variance of the estimator we find for a particular design to that for the optimal design.

As an example, we see the efficiency of the designs found for a particular network, again using Example 2 from Parker et al. [5]. As before, we arbitrarily chose point priors of $\beta_0 = \mu = 0 = 2, \beta_1 = \tau_1 = 1, \beta_2 = \tau_2 = 0, \sigma^2 = 1$.

We plot the efficiency for all designs as Figure 6. In this plot, the maximum possible efficiency (if we use the optimal design) is 1. We show the efficiencies for all possible designs (where we do not insist that we have balanced treatments, an equal number of treatment 1 and treatment 2), and those where we insist on balanced designs, which would be the natural assumption if we did not know there was a network effect.

The plot shows that efficiency is generally much higher for ψ_τ , estimating the treatment effects, than for ψ_ρ . The optimal design for ψ_τ is often balanced, so both plots for ψ_τ are similar, but the median efficiency is around 0.95. We see efficiencies as low as 0.25, however.

For ψ_ρ , efficiency is often really poor. The optimal design is not balanced, so if we choose a balanced design at random, the efficiency is typically (at the median) around 0.35. Choosing a design with the same treatment allocation as the optimal does a little better, but median efficiency is still only 0.5.

The clear message is that if we suspect there is an autoregressive effect, we must include it in our model when considering the design. For this particular prior, the loss in efficiency in estimating ρ effects is often small, but can be very large in the worst case. For estimating τ there is also a very important message: if there is a network design, we often produce highly efficient designs by chance, but there is a good chance that the design will be very inefficient. We therefore waste resources or run the risk of producing estimates with high variance if we do not take into account an autoregressive effect where it exists.

Figure 7 shows, for the same network and prior, the two information criteria ψ_ρ and ψ_τ for three arbitrary balanced designs. Here we see how the information criteria values change when we assume the model, for different values of unknown ρ , $0 < \rho < 1$. This shows clearly that the information criterion ψ_τ varies substantially for this network depending on the (usually unknown) value of ρ , but that for ψ_ρ varies less. Different designs give lower variances for different values of ρ . Considering ψ_τ for estimating

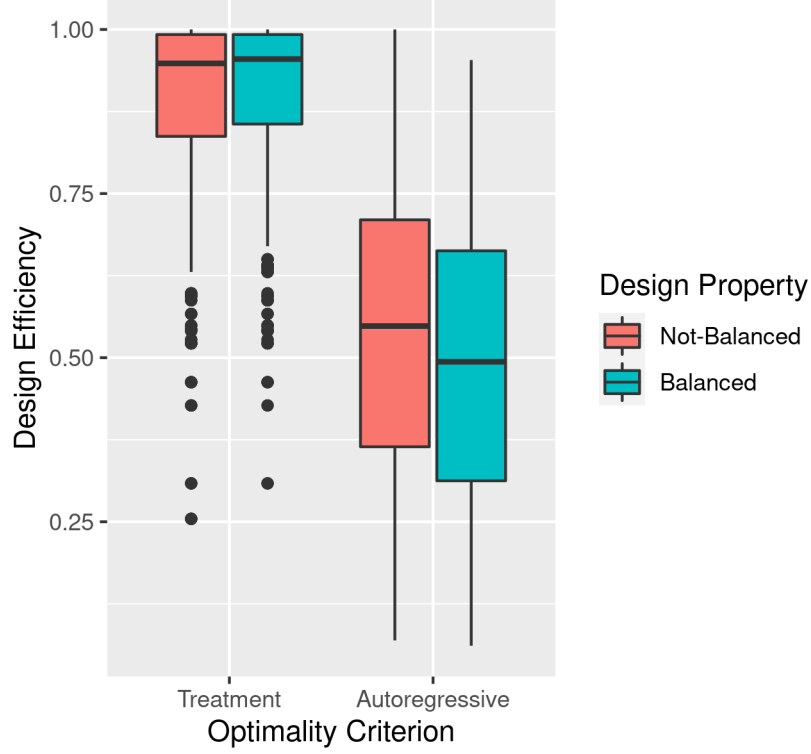


Fig. 6: Efficiency of designs. Left pair of plots shows efficiency for ψ_τ , right pair of plots the efficiency for ψ_ρ . Within each pair, the red plot shows the efficiency assuming that the design has the same number of plots allocated to each treatment as the optimal design, whereas the blue plot assumes that a balanced design is chosen, where each treatment equally replicated.

the treatment effects the red design is usually better, except for values between 0.581 and 0.650 when blue is slightly better, and between 0.664 and 0.734 when purple is slightly better. We have shown just three arbitrary designs for clarity, but the general pattern is true for other designs.

The conclusion is the same: for some values of ρ there are clear differences in the lines, meaning that there is potentially a large difference in efficiencies between designs when there is a network effect.

4.5 Spatial Experiment Example

We now describe a practical experiment that could be used within this framework. It is necessarily a conceptual example, but demonstrates the versatility and importance of this method.

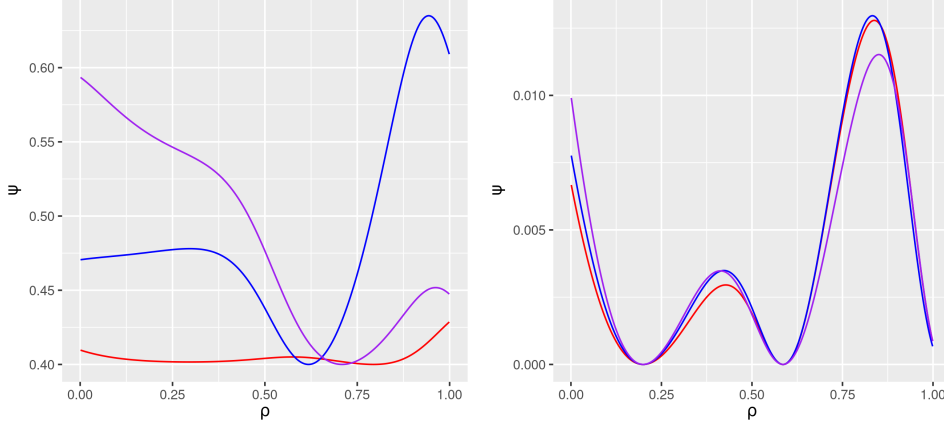


Fig. 7: Optimality function values for (left) ψ_τ and (right) ψ_ρ for varying ρ for three arbitrary balanced designs: $(1,1,1,1,1,2,2,2,2,2)$ in red, $(1,2,1,2,1,2,1,2,1,2)$ in blue, and $(1,1,2,2,1,1,2,2,1,2)$ in purple.

A London Borough (we use Hillingdon as an example) consists of a number of constituent Wards, and the local council want to assess the effectiveness of an intervention which promotes recycling by advertising in some wards. It is thought that promoting recycling in one ward might have an effect on recycling in other wards, as residents move between wards. The response measured is the weight of waste recycled per household. Treatment 1 is to advertise the recycling scheme to all households in the ward by means of a billboards in each ward, whereas treatment 2 is not to do this.

Hillingdon is a varied borough, and it is often thought there is a split between the relatively wealthy North of the borough, and the less wealthy South. It might therefore be of interest to divide the borough into two blocks, in case wealth affects the amount recycled.

It could be reasonable to assume that the effect of an advertising campaign might work in two ways: firstly, people travelling from nearby wards might see the billboards in other wards and their behaviour altered. Secondly, increasing recycling in one area might promote more recycling in other areas through community internet sites, etc. Model (1), or the full model (2) adding blocks and indirect treatment effects might both be appropriate.

We let the adjacency matrix be $A_{ij} = 1/d(i, j)$, $i \neq j$ where $d(i, j)$ is the distance between the geographical centroids of the wards, and $A_{ii} = 0$. We adopt the point priors $\beta_0 = \mu = 20$, $\beta_1 = \tau_1 = 10$, $\beta_2 = \tau_2 = 0$, $\sigma^2 = 1$ and present results for $\rho = 0.1$.

We consider the optimal design found under both model (1) and the full model (2). Due to the size of the network, these are near optimal designs found using an Exchange algorithm on the network with 10000 iterations. The block structure and the optimal designs are shown in Figure 8.

We see that the designs are quite similar in a lot of places, although there are some differences. We present as Figure 9 the efficiencies of choosing a random design (both balanced and non-balanced) compared to the optimal design found.

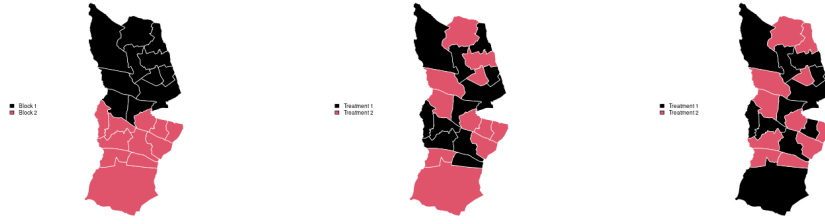


Fig. 8: For the Experiment on the London Borough of Hillingdon, we show (left) the block structure dividing the borough into North and South (center) the optimal design under model (1) (right) the optimal design under the full model (2)

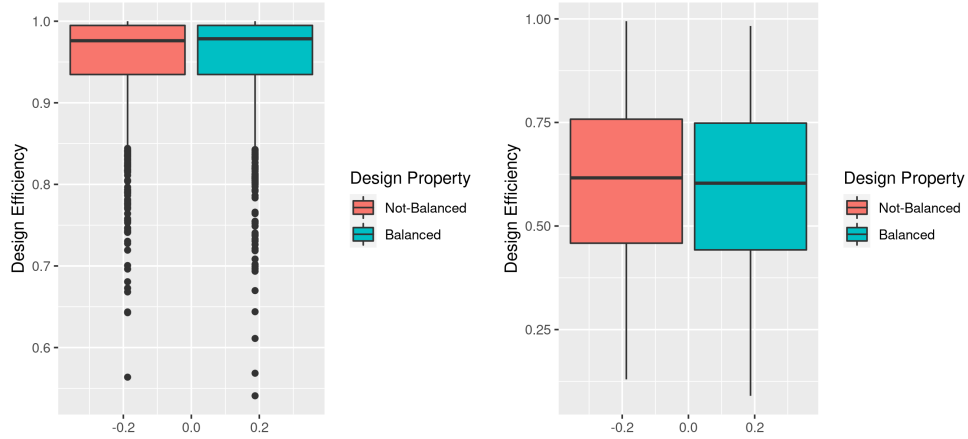


Fig. 9: The efficiencies for 1000 randomly chosen balanced and non-balanced designs found under (left) model (1) and (right) the full model (2)

We see that it is crucial to choose the right design; for model (1), although a random treatment allocation often does not do too badly, there are some designs which have efficiency close to 0.5. If there is a network effect in the model, and we do not include one, we might thus waste almost half the experimental resources. With the full model, which includes indirect treatment effects, the result is even more pronounced- again, if there are indirect treatment effects they must be included in the network, or the median efficiency is only just over 60%.

4.6 Further experiments

Optimal designs for further experiments are shown in the Appendix.

5 Conclusions

This work provides a clear methodology for designing experiments for autoregressive models. We have provided a methodology for finding designs for a wide range of networks, for example social networks with binary adjacency matrices, and spatial networks where distances are known, so this methodology could be useful across a wide range of experiments. Both directed and undirected networks are covered in this methodology.

The efficiency plots show how vital it is to account for the autoregressive parameter if indeed it is present; not accounting for an autoregressive design, and wrongly choosing a randomised balanced design could have large practical effects for an experiment, resulting in potentially a very large loss in efficiency.

We have provided software which allows optimal designs to be found, given a network and a prior. Our software finds exact designs for small networks with a small number of treatments, given priors on the treatment and autoregressive parameters, although combined with algorithms described in [11] near optimal designs for larger networks could be found. We intend to develop these algorithms in the future to allow optimal, or near-optimal designs to be found for much larger networks.

5.1 Future work

In this paper we investigated autoregressive effects, but explained how we could expand our model to incorporate indirect treatment effects. We intend to develop this in future work.

There are many further models that could be used. For example, our models implicitly assume that the treatments are given, and that the responses of our network of experimental units all simultaneously change their responses in line with their neighbours such that the correlation structure is obeyed. This might be realistic for an agricultural experiment, but in a social network it may be that responses are revealed sequentially; a directed network could help in this regard, but the exact mechanism for gathering the responses may change the model and thus the design significantly.

We also implicitly assume in model (2) that the way that indirect treatment effects and autoregressive effects propagate through the network is the same, in that it is specified by the adjacency matrix A . We could generalise this model so that these effects propagate in different ways, e.g.

$$Y_{ij} = \mu + \tau_{t(i)} + z_{b(i)} + \sum_{k=1}^n A_{ki} \gamma_{t(k)} + \rho \sum_{k=1}^n A'_{ki} Y_k + \epsilon_{ij}, \quad (4)$$

where A' is an adjacency matrix such that $A' \neq A$.

We do not discuss methods of estimation here based on the designs we found. There has been much research on how to analyse the MRSAR model (our model (1)), for example numerical (two-step) MLE or using a generalised method of moments. For an overview of estimation see [17], where he says “there is no satisfactory two-step procedure and estimators for the parameters must be obtained from an explicit maximization of the likelihood”. How the analysis technique impinges on a design

is not considered in this paper (we assume an MLE is found), although should be addressed in future work.

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Appendix

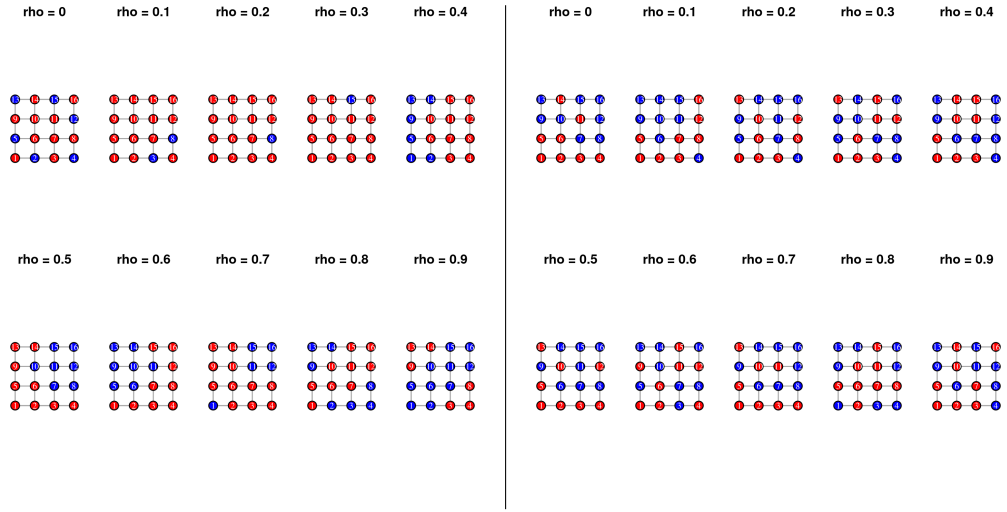


Fig. 10: Optimal design for two treatments on 4×4 lattice network network for estimating (left) ψ_ρ and (right) ψ_τ

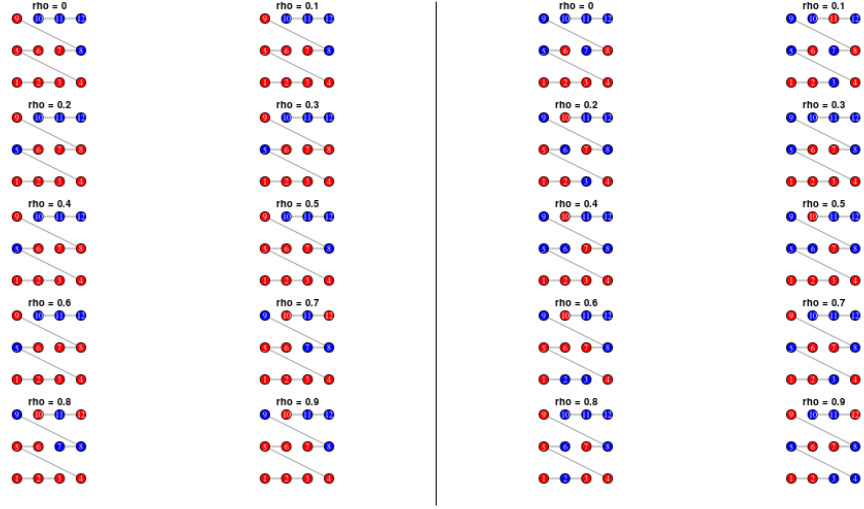


Fig. 11: Optimal design for two treatments on example 3 network from [5] for estimating (left) ψ_ρ and (right) ψ_τ

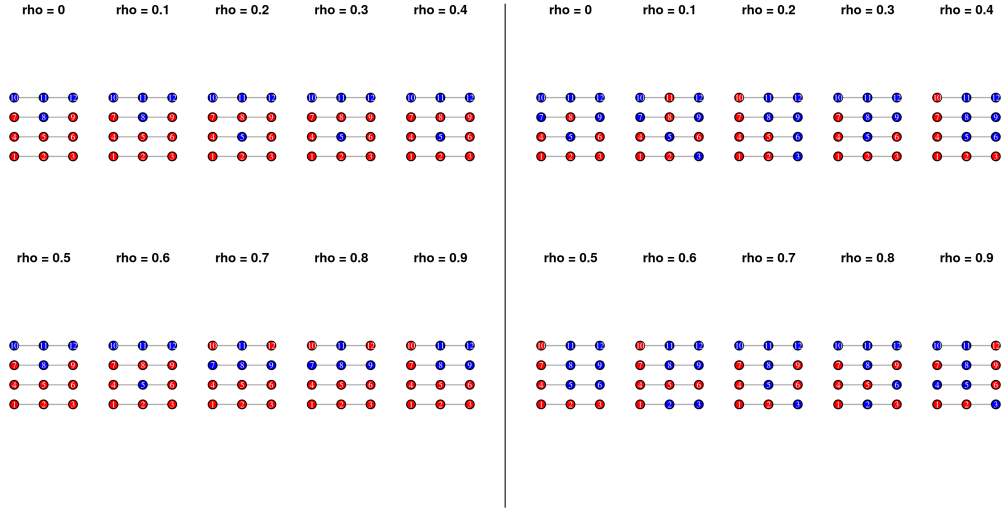


Fig. 12: Optimal design for two treatments on example 4 network from [5] for estimating (left) ψ_ρ and (right) ψ_τ

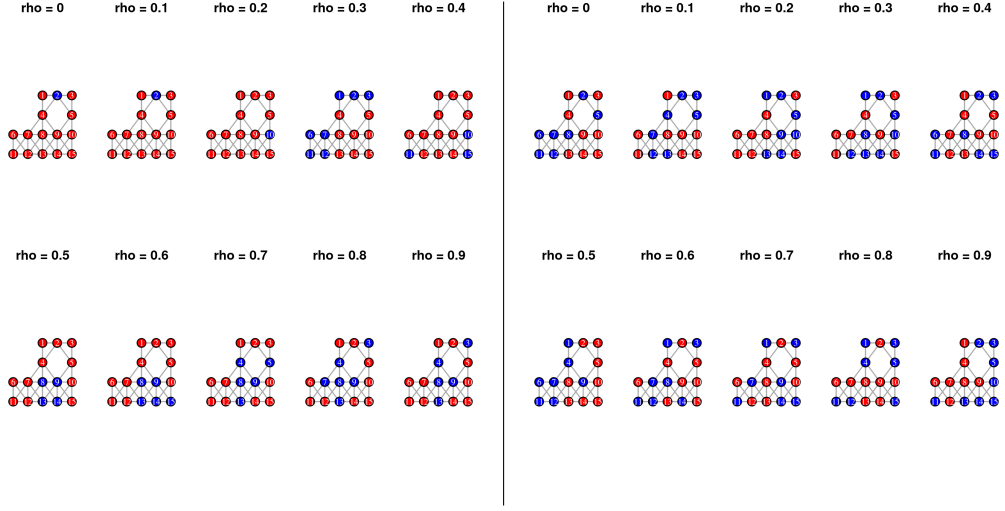


Fig. 13: Optimal design for two treatments on example 5 network from [5] for estimating (left) ψ_ρ and (right) ψ_τ

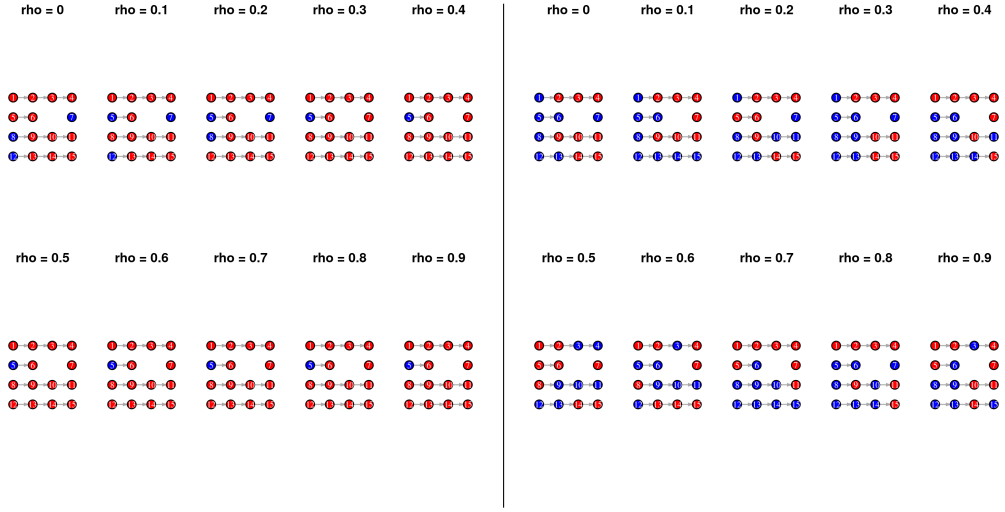


Fig. 14: Optimal design for two treatments on example 6 network from [5] for estimating (left) ψ_ρ and (right) ψ_τ